

Incident report INC9344

Severity: critical. Service: auth-service. Commander: Amara Novak (incident commander).

Opened 2024-04-24T02:10:00Z, resolved 2024-04-24T03:58:00Z. This report is the authoritative record for INC9344.

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Scope and blast radius

Card and wallet authorisations routed through payment-gateway failed for approximately 131 minutes in eu-central-1.

The reference failure raised by the customer is Txn 83155, raised by Verdemar Holdings (Customer 38198).

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Timeline of record

01:12 UTC - release 2.14.0 of auth-service reaches production.

01:47 UTC - token validation p99 crosses 8 seconds.

02:03 UTC - TX83155 fails with `upstream_auth_timeout`.

02:10 UTC - INC9344 opened by Helena Okonkwo.

03:25 UTC - release 2.14.0 rolled back.

03:58 UTC - authorisation success rate returns to baseline.

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Customer impact

Verdemar Holdings reported a declined payment of 5,053.22 EUR. The payment reference quoted on the support ticket was payment reference 83155.

Twelve further authorisations from other merchants failed inside the same window with the identical upstream code, which is why this was treated as a platform incident and not a single-merchant dispute.

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Telemetry: auth-service

Token validation p99 rose from 220ms to 8.4s within 35 minutes of the release. Thread pool saturation is visible in the same window.

The regression was introduced by a synchronous revocation-list lookup added in 2.14.0; the lookup was not cached and contended on session-db.

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Telemetry: payment-gateway

payment-gateway holds an authorisation session open until auth-service answers. With answers arriving after 8 seconds, the 64-connection pool was fully occupied.

Once the pool was exhausted the gateway rejected new authorisations upstream, which is the code the merchant saw.

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Telemetry: ledger-service

ledger-service recorded no write errors during the window. No funds moved for the failed authorisations, so no reconciliation was required.

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Hypothesis 1 - fraud rule rejection

A velocity rule shipped to fraud-service at 16:12 UTC on 23 April 2024 was the first suspect, and INC1719 was raised for it.

The rule did block nine unrelated high-value card payments. It did not block TX83155: the fraud-service decision log records ALLOW for that authorisation.

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Hypothesis 2 - network failure

INC7512 describes transit packet loss in eu-central-1. That event is dated 20 April 2024, three days before this incident.

Interface counters for the incident window show no loss, so the network hypothesis does not survive the timeline.

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Hypothesis 3 - insufficient balance

The account behind C38198 had an available balance of 18,400.00 EUR at the time of the attempt, against an authorisation of 5,053.22 EUR.

The authorisation also never reached the issuer, so an issuer-side decline was not possible.

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Evidence inventory

Gateway access logs, auth-service traces, the fraud decision log, interface counters, the customer statement and the payments dashboard screenshot were all reviewed.

The dashboard screenshot captured at 02:10 UTC shows TX83155 in the failed state.

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Change management

The rollback of 2.14.0 was executed as an emergency change. The approval record for that change is held in the incident approval memo for INC9344; version 2 of that memo is the current one.

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Contributing factors

No canary stage existed for the revocation-list lookup, and the gateway had no bounded wait on auth-service. Either control alone would have contained the incident.

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Root cause determination

Root cause confirmed: auth-service token validation p99 rose to 8.4s after release 2.14.0, so payment-gateway held authorisation sessions open until its 64-connection pool was exhausted. TX83155 was declined with `upstream_auth_timeout` because the authorisation call never returned, not because of fraud scoring, network loss or an insufficient balance.

Confidence: high. The determination is supported by the auth-service trace sample, the payment-gateway pool metric, the fraud-service ALLOW decision for TX83155 and the customer statement.

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Remediation

Cache the revocation list with a 30-second TTL, add a 1.5-second bounded wait in payment-gateway, and stage the next auth-service release behind a canary.

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Follow-up actions

ACT-1 owner Marcus Halvorsen: ship the cache. ACT-2 owner Ruben Lindqvist: bounded wait. ACT-3 owner Helena Okonkwo: canary stage. All actions tracked against INC9344.

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